

AI Lab Final Project Report

Protein Detection and Annotation Using Foundation Models and Retrieval-Augmented Systems

Arybella Theul & Omar Dovesi

Student Lead: Aidan McCrillis

Advisor: Professor Zhang

University of Rhode Island AI Lab

1. Introduction

This project explores the use of foundation models in computational biology to improve protein detection and annotation, with a specific focus on identifying plastic-degrading proteins. The work integrates machine learning, bioinformatics, and data infrastructure to analyze protein sequences and enhance model interpretability.

The primary objective of the project is to compare two approaches to protein annotation:

- A pipeline-based reference dataset (PDG-Finder)
- A transformer-based protein language model (Gaia / gLM2)

Rather than simply comparing performance, this project aims to **enhance and support model outputs** through the integration of a **Retrieval-Augmented Generation (RAG) system**, enabling biologically grounded interpretation of predictions.

2. Background and Motivation

Plastic pollution is a global environmental issue, and identifying proteins capable of degrading plastics is an important step toward sustainable solutions. Traditional bioinformatics approaches rely on sequence similarity and curated databases, but recent advances in **protein language models** allow for deeper representation learning.

The project is specifically focused on improving prediction of plastic-degrading genes based on:

- Protein structure and folding
- Active sites of biochemical reactions
- Genomic context

This motivates the need to:

- Compare model representations
- Improve annotation accuracy

- Provide biological context to predictions

3. Project Overview

The project consists of three main components:

1. Model Comparison

- Compare embeddings from:
 - Gaia model
 - Pipeline reference dataset (PDG-Finder)
- Use cosine similarity and clip scores to evaluate similarity

2. Embedding Analysis

- Generate embeddings for protein sequences
- Compare:
 - Target protein vs. Gaia reference
 - Target protein vs. pipeline reference

3. RAG Database System

- Build a system that:
 - Stores embeddings
 - Retrieves biologically relevant matches
 - Enhances model outputs with contextual information

4. Work Completed

4.1 Foundational Learning (Month 1)

We began by building foundational knowledge in both biology and machine learning:

- Studied molecular biology concepts (protein structure, folding, function)
- Reviewed key papers:
 - Gaia paper
 - OMG dataset
 - Protein foundation models
- Learned how protein language models generate embeddings
- Understood project goals through meetings with Aidan and Professor Zhang

These meetings helped clarify that the goal is not just comparison, but **enhancing and supporting results from both models**.

4.2 Model Understanding and Pipeline Exposure

Through the project, we gained exposure to:

- gLM2 model architecture (transformer-based encoder)
- Embedding generation workflows
- HPC environments (Unity cluster, rsync, scratch space)

We also learned how:

- Embeddings are compared using cosine similarity
- Clip scores are used to evaluate similarity between sequences
- Clusters of proteins are analyzed for functional similarity

4.3 Analysis of Model Outputs

Aidan analyzed differences between Gaia and pipeline outputs using:

- Cosine similarity comparisons
- Clip score distributions
- Cluster-based analysis

These analyses helped identify:

- Where each model performs better
- Potential biases in reference datasets
- Differences in how embeddings capture biological signals

4.4 Transition to RAG System (Month 2)

Following discussions with the team, the project shifted toward building a **Retrieval-Augmented Generation (RAG) system**.

This idea was introduced as an alternative to traditional vector databases:

- Instead of just storing embeddings,
- Use retrieval to **augment and support predictions with biological context**

4.5 RAG Database Development

We began implementing the RAG-based system.

Arybella:

- Built a **toy vector database using Chroma in Google Colab** to simulate the retrieval pipeline
- Loaded protein sequence data and generated embeddings using provided datasets

- Structured and tested a **basic embedding → storage → retrieval workflow**
- Performed similarity-based queries to validate that nearest-neighbor retrieval was functioning correctly

Google Colab Rag notebook

- Initial database built using Chroma for small datasets
- Plan to scale depending on memory (<16GB preferred)
- Explored alternatives:
 - FAISS (open-source)
 - QDrant (hosted solution)

4.6 Database Design Decisions

Important design decisions included:

- Merging Gaia + pipeline datasets into one system
- Adding labels to distinguish dataset origin
- Using embedding similarity for retrieval
- Structuring data for scalability

These decisions ensure that:

- Model outputs can be directly compared
- Retrieval results provide meaningful biological context

4.7 Scaling and Performance Considerations

Omar:

- Executed large-scale embedding generation for protein sequences using the HPC environment
- Observed that embedding generation is computationally expensive:
 - ~2 hours for 100,000 sequences
 - Estimated ~70 days for the full dataset
- Demonstrated that once embeddings are generated, **retrieval operations are highly efficient and fast**
- Contributed to scaling considerations for integrating embeddings into the database system

4.8 Individual Contributions

Arybella:

- Focused on understanding the integration of protein language models with retrieval systems
- Developed and tested an initial RAG-based database using Chroma
- Validated embedding outputs and retrieval behavior on small datasets

- Contributed to early-stage system design and pipeline structure

Omar:

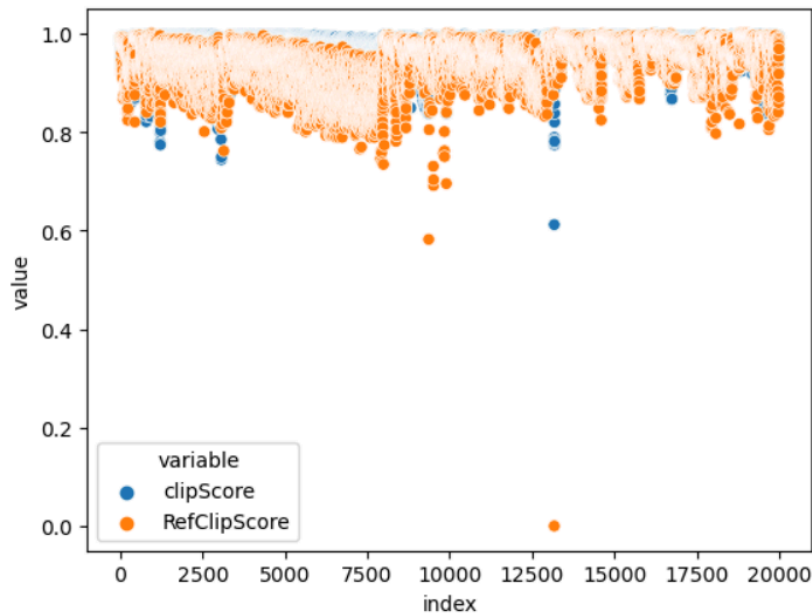
- Focused on large-scale embedding generation and system scalability
- Ran embedding scripts on HPC infrastructure
- Evaluated performance constraints and runtime requirements
- Contributed to database scaling and integration planning

5. Key Results and Insights

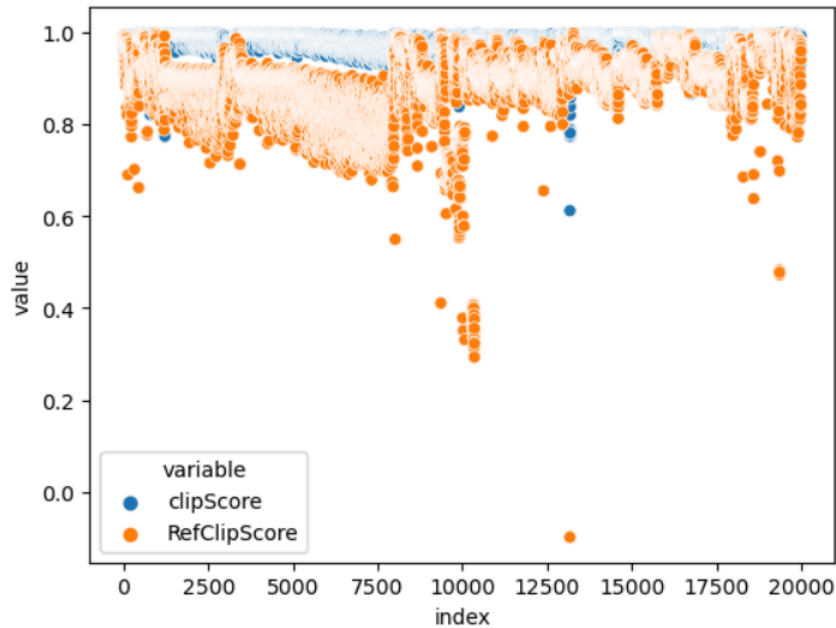
From the work completed:

- Gaia embeddings often produce higher similarity scores than pipeline references
- Some discrepancies exist due to:
 - Averaging vs. max similarity calculations
 - Differences in embedding generation methods
- Cluster-level analysis shows variability across protein types

New Distribution of clip scores



Old Distribution of clip scores



Additionally:

- RAG system enables **interpretability**, not just prediction
- Combining datasets improves analytical flexibility

6. Challenges Encountered

- Difficulty matching Gaia's exact clip score calculations
- Computational cost of embedding large datasets
- Designing a scalable database architecture
- Interpreting embeddings biologically

7. Resources and Materials Created

To centralize all work, the following resources were created:

Code & Notebooks

- RAG toy database (Google Colab)
- Embedding generation scripts
- Retrieval testing scripts

[Colab Notebook](#)

Datasets

- Gaia dataset (OG_Prot_90)
- Pipeline reference dataset
- Narragansett Bay dataset

Documentation

- Notes on Gaia vs pipeline comparison
- Apptainer setup documentation
- HPC workflow notes

[Meeting Notes](#)

9. Conclusion

This project demonstrates how machine learning models can be enhanced through structured retrieval systems to improve biological interpretation.

This work bridges machine learning and biology, contributing toward more interpretable and scalable protein detection systems.